Light

Network Hardware & Topologies

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Edexcel GCSE Computer Science — **Network Hardware and Topologies (1CP2/01 Topic 4: Networks)**. This topic covers why computers are networked, the difference between LANs and WANs, how the internet is structured, network hardware, wired and wireless connectivity characteristics, and the three network topologies you need to know. Exam questions typically ask you to **identify** network types and characteristics, **describe** the role of hardware components, **compare** topologies, and **construct expressions** for transmission rate calculations. Expect questions worth 1—4 marks requiring clear definitions and comparisons.

Why networks matter

Networks are fundamental to modern computing. Almost every device we use connects to a network, from smartphones to smart home devices to business servers.

- **Communication:** Networks enable email, video calls, instant messaging, and social media
- **Resource sharing:** Multiple users can share printers, storage, and internet connections
- **Data access:** Files and databases can be accessed from multiple locations

- **The internet:** The world's largest network connects billions of devices globally
- **Paper 1 Topic 4:** Network questions appear regularly, often worth 10—14 marks total

What you must know

This checklist covers the key exam-visible concepts from Edexcel 1CP2 Topic 4: Networks (specifically 4.1.1—4.1.5 and 4.1.8).

- **Understand** why computers are connected in a network
- **Understand** different types of networks (LAN, WAN)
- **Understand** how the internet is structured (IP addressing, routers)
- **Understand** how wired and wireless connectivity characteristics impact performance (speed, range, latency, bandwidth)
- **Understand** that network speeds are measured in bits per second (kilobit, megabit, gigabit)
- **Construct** expressions involving file size, transmission rate, and time
- **Understand** characteristics of network topologies (bus, star, mesh)

> **Specification link:** This topic covers Edexcel 1CP2 Paper 1 Topic 4: Networks — specifically 4.1.1 (why networks), 4.1.2 (LAN/WAN), 4.1.3 (internet structure), 4.1.4 (wired/wireless characteristics), 4.1.5 (network speeds and expressions), and 4.1.8 (topologies).

Toolkit

These are the key formulas, patterns, and reference tables you need for network questions. Memorise the transmission time formula and the topology characteristics — they appear frequently.

Network speed units

Unit	Abbreviation	Value in bits per second	Typical use
Bit per second	bps	1	Base unit
Kilobit per second	kbps / kb/s	1,000 (10^3)	Slow connections, audio streaming
Megabit per second	Mbps / Mb/s	1,000,000 (10^6)	Home broadband, video streaming
Gigabit per second	Gbps / Gb/s	1,000,000,000 (10^9)	Fast fibre, data centres

> **Important:** Network speeds use **decimal** prefixes (1000), not binary prefixes (1024). This is different from file storage units! 1 Mbps = 1,000,000 bits per second.

Transmission time formula

Time (seconds) = File size (bits) ÷ Transmission rate (bits per second)

Or rearranged:

Rate = File size ÷ Time | **File size = Rate × Time**

Unit conversion reminder: Before using the formula, ensure all units match. If file size is in bytes, multiply by 8 to get bits. If file size is in gibibytes, convert to bytes first ($\times 1024^3$), then to bits ($\times 8$).

Topology comparison at a glance

Topology	Structure	Key advantage	Key disadvantage
Bus	All devices on single backbone cable	Simple, cheap to install	Single point of failure (backbone)
Star	All devices connect to central switch/hub	One device failure doesn't affect others	Central device is single point of failure
Mesh	Devices connected to multiple other devices	Highly reliable, multiple paths	Expensive, complex to set up

Wired vs wireless at a glance

Characteristic	Wired (Ethernet)	Wireless (Wi-Fi)
Speed	Generally faster (up to 10 Gbps)	Generally slower (varies with distance)
Range	Limited by cable length (100m for Ethernet)	Limited by signal strength (typically 30—50m indoors)
Latency	Lower (more consistent)	Higher (more variable)
Bandwidth	Higher, more stable	Shared, can be congested
Interference	Minimal	Affected by walls, other devices

1. Network fundamentals

A **network** is two or more computers or devices connected together so they can communicate and share resources. Networks range from small home setups to the global internet.

1.1 Why computers are networked

There are many benefits to connecting computers in a network:

- **File sharing:** Users can access files stored on a central server or on other computers, making collaboration easier

- **Hardware sharing:** Expensive resources like printers, scanners, and storage can be shared between multiple users
- **Internet access:** A single internet connection can be shared by many devices through a router
- **Communication:** Networks enable email, instant messaging, video conferencing, and VoIP calls
- **Centralised backup:** Data can be backed up to a central location, making it easier to manage and protect
- **Software distribution:** Programs and updates can be installed across multiple computers from a central point
- **Remote access:** Users can access their files and applications from different locations

1.2 Types of networks: LAN and WAN

Networks are classified by their geographical size. At GCSE, you need to know two main types:

LAN – Local Area Network

A network covering a **small geographical area**, typically a single building or site.

- Examples: home network, school network, office network
- Usually owned and managed by a single organisation
- Uses private infrastructure (cables, switches, routers)
- Relatively fast data transfer speeds
- Low cost to maintain

WAN – Wide Area Network

A network covering a **large geographical area**, connecting LANs across cities, countries, or continents.

- Examples: the internet, a company's global network
- Often uses third-party infrastructure (telephone lines, satellite links)
- Can connect LANs that are far apart
- Generally slower than LANs due to distance
- Higher cost due to renting infrastructure

Key differences between LAN and WAN:

Feature	LAN	WAN
Geographic size	Small (single building/site)	Large (cities, countries, global)
Ownership	Usually owned by one organisation	Often uses third-party infrastructure
Speed	Faster (shorter distances)	Slower (longer distances)
Cost	Lower (own infrastructure)	Higher (renting connections)

Feature	LAN	WAN
Example	School network, home network	The internet, corporate global network

1.3 Network speeds

Network speeds measure how fast data can be transmitted. Speeds are measured in **bits per second** (bps), not bytes per second.

> **Key distinction:** Network speeds use **bits**. File sizes typically use **bytes**. Since 1 byte = 8 bits, you must multiply file size in bytes by 8 when calculating transmission times.

Common speed measurements:

- **Kilobit per second (kbps):** 1,000 bits per second — used for older dial-up connections
- **Megabit per second (Mbps):** 1,000,000 bits per second — typical home broadband
- **Gigabit per second (Gbps):** 1,000,000,000 bits per second — fast fibre connections

1.4 Calculating transmission time

Exam questions often ask you to construct expressions for transmission time. The key formula is:

> **Time = File size ÷ Transmission rate**
> Ensure units are consistent before dividing.

Example conversion steps:

1. Convert file size to bits (if in bytes: ×8)
2. Convert file size to the same unit prefix as the rate
3. Divide to get time in seconds
4. Convert seconds to minutes if required (÷60)

Common unit conversions for exam questions:

- **Gibibytes to bytes:** Multiply by 1024^3 (or 1024 × 1024 × 1024)
- **Bytes to bits:** Multiply by 8
- **Minutes to seconds:** Multiply by 60
- **Bits to megabits:** Divide by 1000^2 (or 1,000,000)

Understanding why units matter: The transmission rate tells you how many bits are sent per second. If your file size is in bytes, you must convert to bits first. Similarly, if the rate is in Mbps and you want the answer in megabits, you need to ensure the file size is also in megabits.

> **Exam tip:** When asked to "construct an expression", show all your working with clear unit conversions. You often don't need to calculate the final numerical answer — just set up the expression correctly. Each conversion step can earn marks.

> Worked Example 1 – Identifying network type

Question: A school has computers in multiple classrooms, a server room, and a staff room, all connected together. What type of network is this? [1 mark]

Model answer:
LAN / Local Area Network **(1)**
Reasoning: A school is a single site/building — this is a small geographical area, so it's a LAN.

Exam tip: Focus on geographical size. Single building/site = LAN. Multiple sites/cities = WAN.

Worked Example 2 — Benefits of networking

Question: Give two benefits of connecting computers in a network. [2 marks]

Model answer:
1. Resources such as printers can be shared between users **(1)**
2. Files can be accessed from any computer on the network **(1)**

Exam tip: Give specific benefits, not vague answers. "Sharing" alone is not enough — say what is shared (files, printers, internet connection).

Worked Example 3 — Network speed units

Question: A home broadband connection has a speed of 100 Mbps. Express this speed in bits per second. [1 mark]

Model answer:
100,000,000 bits per second / 100×10^6 bps **(1)**
Working: 100 Mbps = 100 × 1,000,000 = 100,000,000 bps

Exam tip: Remember: 1 Mbps = 1,000,000 bps (using decimal prefixes for network speeds).

Worked Example 4 — Simple transmission time

Question: A 50 megabit file is transmitted over a 10 Mbps connection. Calculate the transmission time. [2 marks]

Model answer:

Step 1: Identify values: File size = 50 Mb, Rate = 10 Mbps **(1)**
Step 2: Apply formula: Time = 50 ÷ 10 = 5 seconds **(1)**

Exam tip: This question gave the file size in megabits, matching the rate unit. Always check if conversion is needed!

2. Network hardware and connectivity

Networks require hardware to function. This includes devices that direct traffic, connect computers, and provide addresses for communication. Understanding how the internet is structured and the differences between wired and wireless connections is essential.

2.1 How the internet is structured

The **internet** is a global network of interconnected networks. It relies on two key concepts to function:

IP Addressing

Every device on the internet has a unique **IP address** (Internet Protocol address) that identifies it.

- Works like a postal address for data packets
- IPv4 format: four numbers 0—255, e.g., 192.168.1.1
- IPv6: newer format with more addresses available
- Public IP: visible on the internet
- Private IP: used within LANs

Routers

A **router** is a device that forwards data packets between networks.

- Connects different networks together (e.g., LAN to internet)
- Uses IP addresses to determine the best path for data
- Maintains routing tables to direct traffic
- Home routers often combine router, switch, and wireless access point

When you send data over the internet, it travels through multiple routers, each one passing the data closer to its destination based on the IP address. This process is called **routing**, and it happens automatically — you don't need to know the path your data takes.

Internet structure: Routers (R1, R2) connect LANs using IP addresses

How data travels across the internet:

1. Your device sends data with the destination IP address attached
2. Your router forwards it to your Internet Service Provider (ISP)
3. The ISP's routers check their routing tables to find the best path
4. Data passes through multiple routers, each one getting closer to the destination
5. The final router delivers the data to the destination device

2.2 Other network hardware

While routers are explicitly in the specification, understanding other common network devices helps with context:

- **Switch:** Connects devices within a LAN, forwards data only to the intended recipient
- **Network Interface Card (NIC):** Hardware that allows a device to connect to a network (wired or wireless)
- **Wireless Access Point (WAP):** Allows wireless devices to connect to a wired network

2.3 Wired connectivity

Wired connections use physical cables to connect devices. The most common type is **Ethernet**, which uses copper or fibre optic cables.

Characteristics of wired connections:

- **Speed:** Generally faster and more consistent — Ethernet supports up to 10 Gbps
- **Range:** Limited by cable length — standard Ethernet works up to 100 metres
- **Latency:** Lower and more consistent delay — better for gaming and video calls
- **Bandwidth:** Higher capacity, less affected by congestion
- **Reliability:** Not affected by interference from other devices
- **Security:** More secure — physical access required to intercept

2.4 Wireless connectivity

Wireless connections use radio waves to transmit data. **Wi-Fi** is the most common wireless technology for networks. Wireless networks have become increasingly popular due to the convenience of not needing physical cables.

Characteristics of wireless connections:

- **Speed:** Generally slower than wired, varies with distance and interference
- **Range:** Measured in metres — typically 30—50m indoors, affected by walls and obstacles
- **Latency:** Higher and more variable — can fluctuate with signal strength
- **Bandwidth:** Shared between all connected devices, can become congested
- **Interference:** Affected by walls, other Wi-Fi networks, microwaves, and other electronic devices
- **Mobility:** Allows devices to move freely within range

Factors affecting wireless performance:

- **Distance from router:** Signal strength decreases with distance
- **Physical obstacles:** Walls, floors, and furniture reduce signal strength
- **Interference:** Other wireless devices and networks can cause interference
- **Number of users:** Bandwidth is shared, so more users means slower speeds for each
- **Frequency band:** 5GHz is faster but has shorter range than 2.4GHz

Exam tip: When asked about wireless characteristics, "range" is measured in **metres** — this is a common exam question! Remember that Wi-Fi signals can be blocked by walls and interfered with by other electronic devices.

2.5 Performance characteristics explained

Characteristic	Definition	Impact on user experience
Speed	How fast data can be transferred, measured in bits per second	Faster downloads, smoother streaming
Range	The maximum distance over which a connection can operate	Determines where devices can be placed
Latency	The delay between sending data and it being received (ping time)	Low latency = responsive gaming, smooth video calls
Bandwidth	The maximum amount of data that can be transmitted at once	Higher bandwidth = more users, bigger files

Worked Example 5 — Role of routers

Question: Describe the role of a router in a network. [2 marks]

Model answer:
A router forwards/directs data packets between different networks **(1)**
It uses IP addresses to determine the best path/route for the data to take **(1)**

Exam tip: Key words for routers: "forwards", "between networks", "IP addresses", "routing". Don't confuse with a switch (which works within a single network).

Worked Example 6 — Wireless characteristic

Question: Name the characteristic of a wireless network that is measured in metres. [1 mark]

Model answer:
Range **(1)**

Exam tip: This is a common exam question. Range = metres. Speed = bits per second. Latency = milliseconds.

Worked Example 7 — Comparing wired and wireless

Question: Give two advantages of using a wired connection instead of a wireless connection. [2 marks]

Model answer:
1. Wired connections have faster/more consistent speeds **(1)**
2. Wired connections have lower latency / less delay **(1)**

Alternative answers: More secure, not affected by interference, more reliable

Exam tip: Think about the four characteristics: speed, range, latency, bandwidth. Wired wins on speed, latency, and bandwidth. Wireless wins on range/mobility.

Worked Example 8 — IP addressing

Question: Explain why every device on the internet needs an IP address. [2 marks]

Model answer:
An IP address uniquely identifies each device on the network **(1)**
This allows data packets to be routed/delivered to the correct destination **(1)**

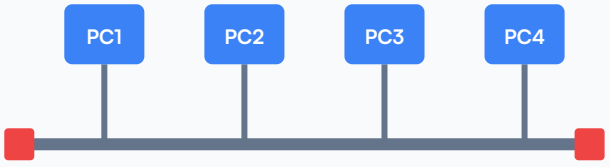
Exam tip: Think of IP addresses like postal addresses — without them, data wouldn't know where to go.

3. Network topologies

A **network topology** describes how devices are arranged and connected in a network. The topology affects performance, reliability, and cost. At GCSE, you need to know three topologies: **bus**, **star**, and **mesh**.

3.1 Bus topology

In a **bus topology**, all devices connect to a single central cable called the **backbone**. Data travels along the backbone in both directions.



Bus topology: all devices on single backbone with terminators (red) at each end

Advantages:

- Simple and cheap to set up
- Uses less cable than other topologies
- Easy to add new devices
- Good for small, temporary networks

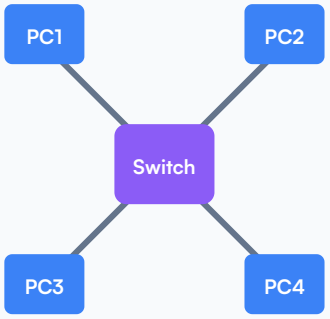
Disadvantages:

- If the backbone fails, the **whole network fails** (single point of failure)
- Performance decreases as more devices are added (collisions)
- Only one device can transmit at a time
- Difficult to troubleshoot faults
- Security risk: all data visible to all devices
- Limited cable length and number of devices

> **Key exam point:** Bus networks require **terminators** at each end of the backbone to prevent signal reflection. If a terminator is missing, the network stops working.

3.2 Star topology

In a **star topology**, all devices connect to a **central switch or hub**. Data passes through the central device to reach its destination.



Star topology: all devices connect to central switch

Advantages:

- If one cable fails, only that device is affected
- Easy to add or remove devices without disruption
- Better performance: no collisions with a switch
- Easy to detect and fix faults
- More secure: data only goes to intended recipient

Disadvantages:

- If the central switch fails, the **whole network fails**
- Requires more cable than bus topology
- More expensive due to switch/hub costs
- Performance depends on central device capacity

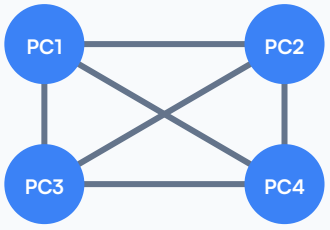
> **Exam tip:** Star is the most common topology in modern networks. Both star and bus have a single point of failure, but they're different: star's is the central switch,

bus's is the backbone cable.

3.3 Mesh topology

In a **mesh topology**, devices are connected to **multiple other devices**. In a **full mesh**, every device connects to every other device. In a **partial mesh**, some devices connect to multiple others but not all devices are directly connected.

How mesh routing works: When data needs to travel from one device to another, the network can choose from multiple available paths. If the direct path is congested or broken, the data automatically takes an alternative route. This is similar to how you might take a different road if your usual route is blocked.



Full mesh: every device connected to every other device

Advantages:

- **Highly reliable:** if one connection fails, data can take another route
- **No single point of failure:** network continues even if devices fail
- Can handle high traffic — multiple simultaneous paths
- Data can take the fastest available route
- Good for critical systems where reliability is essential

Disadvantages:

- **Expensive:** requires many cables and network ports
- **Complex:** difficult to set up and manage
- More difficult to troubleshoot
- Requires more NICs per device in full mesh
- Impractical for large networks (too many connections needed)

Where mesh is used: Mesh topology is common in wireless mesh networks (e.g., home Wi-Fi extenders) and in the internet backbone where reliability is critical. Most LANs use star topology because it balances cost and reliability.

Worked Example 9 — Bus topology disadvantages

Question: Give two disadvantages of a bus network topology. [2 marks]

Model answer:

1. If the backbone cable fails, the whole network stops working **(1)**

2. Performance decreases / more collisions as more devices are added **(1)**
Alternative answers: Security risk (all data visible), only one device can transmit at a time, difficult to troubleshoot

Exam tip: The backbone being a single point of failure is the most important disadvantage of bus topology — it appears frequently in exams.

Worked Example 10 – Comparing topologies

Question: Explain one advantage of a star topology compared to a bus topology. [2 marks]

Model answer:
In a star topology, if one cable fails, only that device is disconnected **(1)**
In a bus topology, a cable failure would affect the whole network **(1)**
Alternative: Star has better performance because there are no data collisions when using a switch.

Exam tip: "Explain" requires you to make a comparison or give a reason. Don't just state facts — show how star is better than bus.

Worked Example 11 – Mesh topology benefit

Question: Explain why mesh topology is highly reliable. [2 marks]

Model answer:
Devices are connected to multiple other devices / there are multiple paths **(1)**
If one connection fails, data can be rerouted through an alternative path **(1)**

Exam tip: "Multiple paths" and "alternative routes" are the key concepts for mesh reliability.

Worked Example 12 – Transmission rate expression

Question: Construct an expression to calculate the transmission rate, in megabits per second, required to transmit a 1.4 gibibyte file in 13 minutes. You do not need to do the calculation. [4 marks]

Model answer:
Step 1: Convert time to seconds: 13 × 60 = 780 seconds **(1)**

Step 2: Convert file size to bits: $1.4 \times 1024^3 \times 8$ **(1)**
Step 3: Convert to megabits: divide by 1000^2 (or 10^6) **(1)**
Step 4: Complete expression:

$$\text{Rate} = (1.4 \times 8 \times 1024^3) \div (13 \times 60 \times 1000^2) \text{ Mbps}$$ **(1)**

Exam tip: This is a 4-mark question requiring careful unit conversion. Remember: gibibytes use 1024, megabits use 1000. Show each conversion step clearly.

Worked examples summary

The table below summarises the 12 worked examples covered in this topic.

#	Topic	Question type	Marks	Key skill
1	Network types	Identify LAN/WAN	1	Geographical size
2	Benefits of networking	Give benefits	2	Specific examples
3	Network speeds	Convert units	1	Mbps to bps
4	Transmission time	Calculate time	2	Apply formula
5	Routers	Describe role	2	Forwarding, IP addresses
6	Wireless characteristic	Identify unit	1	Range = metres
7	Wired vs wireless	Compare advantages	2	Speed, latency
8	IP addressing	Explain purpose	2	Unique identification
9	Bus topology	Give disadvantages	2	Single point of failure
10	Star vs bus	Explain advantage	2	Comparative reasoning
11	Mesh topology	Explain reliability	2	Multiple paths
12	Transmission rate	Construct expression	4	Unit conversion

Quick reference

Key terms and definitions

Term	Definition (mark-scheme style)
Network	Two or more computers/devices connected together to share resources and communicate
LAN	Local Area Network — a network covering a small geographical area such as a single building or site
WAN	Wide Area Network — a network covering a large geographical area, connecting multiple LANs
IP address	A unique numerical identifier assigned to each device on a network
Router	A device that forwards data packets between different networks using IP addresses
Bandwidth	The maximum amount of data that can be transmitted over a connection at once
Latency	The delay between sending data and it being received
Topology	The arrangement/layout of devices and connections in a network
Bus topology	A network layout where all devices connect to a single backbone cable
Star topology	A network layout where all devices connect to a central switch or hub
Mesh topology	A network layout where devices are connected to multiple other devices

Network speed conversions

From	To	Multiply by
Bytes	Bits	× 8
Kibibytes (KiB)	Bytes	× 1024
Mebibytes (MiB)	Bytes	$\times 1024^2$
Gibibytes (GiB)	Bytes	$\times 1024^3$
Megabits (Mb)	Bits	$\times 1000^2$ (or 10^6)
Minutes	Seconds	× 60

Topology comparison table

Feature	Bus	Star	Mesh
Cost	Low	Medium	High
Complexity	Simple	Moderate	Complex
Cable required	Least	More	Most
Single point of failure	Yes (backbone)	Yes (central switch)	No
Reliability	Low	Medium	High
Performance under load	Poor (collisions)	Good	Excellent
Adding devices	Easy	Easy	Difficult
Troubleshooting	Difficult	Easy	Difficult

Typical pitfalls

Pitfall 1 — Confusing bits and bytes in network speeds.
Wrong: "100 Mbps means 100 megabytes per second."
Right: Network speeds use **bits**. 100 Mbps = 100 megabits per second = 12.5 megabytes per second (÷8).

Pitfall 2 — Using wrong prefixes for network speeds.
Wrong: Using 1024 for Mbps conversions.
Right: Network speeds use decimal prefixes: 1 Mbps = 1,000,000 bps. File storage uses binary (1024).

Pitfall 3 — Confusing routers and switches.
Wrong: "A router connects devices within a LAN."
Right: A router connects different networks (e.g., LAN to internet). A switch connects devices within the same network.

Pitfall 4 — Saying star has no single point of failure.
Wrong: "Star topology has no single point of failure."
Right: Star topology's central switch is a single point of failure. If it fails, the whole network fails.

Pitfall 5 — Forgetting range is measured in metres.
Wrong: "Wireless range is measured in Mbps."
Right: Range is measured in **metres**. Speed is measured in Mbps.

Pitfall 6 — Not converting minutes to seconds.
Wrong: Using minutes directly in the transmission formula.
Right: Always convert to seconds (×60) before using the formula. Time must match rate units (per second).

Pitfall 7 — Mixing up LAN and WAN definitions.
Wrong: "A WAN is a small network in one building."
Right: LAN = small area (building). WAN = large area (cities, countries). Think L = Local = Limited area.

Pitfall 8 — Vague answers about networking benefits.
Wrong: "Networks allow sharing."
Right: Be specific: "Networks allow sharing of resources such as printers" or "Files can be accessed from any computer."

Pitfall 9 — Forgetting bus needs terminators.
Wrong: Only mentioning backbone failure for bus disadvantages.
Right: Bus topology also needs terminators at each end — if missing, signals reflect and corrupt data.

Pitfall 10 — Not explaining mesh reliability properly.
Wrong: "Mesh is reliable because it has many connections."
Right: Mesh is reliable because if one path fails, data can be rerouted through alternative paths to reach its destination.

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